

LC 1289 - Min Falling Path Sum II

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here the twist is, we dont simply need min falling path sum but instead:

→ **only allowed to choose one element from each row of grid , st no 2 elements in adjacent rows are in same column**

[THIS IS CALLED ZERO SHIFT]

in falling path sum I →

[r+1, c]
[r+1, c+1]
[r+1, c-1]

were allowed

in falling path sum II →

here literally any other column is allowed except itself.

(straight vertical drop is not allowed,
zigzag/shifting is allowed)



Handwritten notes diagram

Reference: Original Handwritten Visualizations

[at last take min from bottom row]

C++ Implementation

Solution.cpp

```
class Solution {
public:
    int minFallingPathSum(vector<vector<int>>& grid) {

        int n = grid.size();
        int m = grid[0].size();

        vector<vector<int>> dp(n, vector<int>(m, INT_MAX));

        for(int j = 0; j < m; j++){
            dp[0][j] = grid[0][j];
        }

        for(int r = 0; r < n - 1; r++){
            for(int c = 0; c < m; c++){

                int nr = r + 1;
                for(int nc = 0; nc < m; nc++){
                    if(nc == c) continue;
                    dp[nr][nc] = min(dp[nr][nc], dp[r][c] + grid[nr][nc]);
                }
            }
        }

        int ans = INT_MAX;
        for(int col = 0; col < m; col++){
            ans = min(ans, dp[n - 1][col]);
        }
        return ans;
    }
};
```